2019 Placement Interview Experiences @ IIT Kanpur

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About Me:

I was that **unlucky, remember the word "luck",** guy who qualified almost all of the written tests out of those I had written but unable to make the final selection. Later, I got selected in Qualcomm, Hyderabad which is far better than my all previous rejections.
If I'm not wrong, have more interview experiences than anyone else has in SPCOM. Till now, I have forgotten some of the questions but providing you those I remember of and don't worry as soon as I remember the questions, update it. Keep visiting the link. You do not need to worry about interview preparation, I am there to help you. If you need any help you can message me on Facebook https://www.facebook.com/aditya.jaiswal.106). Currently, I am busy with thesis writing so you may face li'l delay in getting a reply. Now, coming to Interview questions…

1. **Qualcomm on Day 1 at 5.30 in the morning**
    - How many bits are required to represent the 2^10 + 2^9 + … +2^0 and 2^-1 + 2^-2 + … + 2^-10?
    - I was given some trapezium and asked to divide in the three or four, not exactly remember, equal parts.
    - We had half an hour of discussion on my thesis.

    Comment:
    I will rate it 7 out of 10 on the basis of the quality of the question they asked in the interview. Overall, it was one of my favorite interviews.

2. **Intel on Day 1 in the morning**
    In Intel, there were multiple panels interviewing parallelly. Firstly, I was sent to some panel and was not comfortable there. So requested to change the panel, and was interviewed there also. Now coming to questions...
    - Proof of MOSFET drain's current equation.
    - How will you explain the charging or discharging of a capacitor to laymen?
    - Explain IR drop and EM?
    - How will you minimize or eliminate them **after the fabrication of the chip**?
    - VLSI Project in detail.
    - What are HTh and LTh MOSFET and where are they used, and give some real examples. *Qureshi sir explained very well in class.*

- How many leakage currents are in the MOSFET, and How will you minimize the leakage current in the MOSFET? *Based on my VLSI project or you can read from Analog VLSI book.*
- Secondary Effect in MOSFET? *Better explained in the Analog VLSI book.*
- Domino Logic? *Digital VLSI Book*
- Difference between static and dynamic logic design? *Well taught by Quershi sir.*
- *SRAM / DRAM* working principle and circuit diagram? *Read everything given in the Digital VLSI book related to SRAM/DRAM.*
- Significance of *Setup and Hold time? Read from VLSI Expert website*

Comment:

I will rate it 1 out of 10 on the basis of the quality of the question they asked in the interview and their selection process.
How much you know about VLSI absolutely doesn't matter here. If you are lucky enough then you will get selected even if you know just ABC of VLSI.
You can't say that I will get selected, even if you have answered 99% of the questions. I think they selected blindly. Ladko tumhara katega…. Don't expect much from Intel otherwise, you will ruin your remaining interviews.

3. **Bajaj Auto**
   They asked such questions that I had never heard of. Expecting some Motor, Generator, and mechanical kinda questions. I answered very few questions and didn't remember any questions.

4. **TSMC**
   It had been a very easy interview. During the first round, I was asked the following.
   - How does CMOS inverter work, Explain?
   - IR Drop, EM?
   - Apne baare me kuch btao?
   - Veg or Non-Veg?
   - Khana bna lete ho, nhi banate ho to Sikh lena?

   You may feel that the last few questions are redundant but they have significance, believe me. You will get this after attending the PPT of TSMC.

   During the 2nd Round, I was asked even easier questions than that of the first round. It was an HR round.

- He kept already drawn different symbols of NMOS, PMOS, CMOS inverter, and I was asked to identify them. Lol…
- Setup time and Hold time violation?
- Make CMOS NAND, NOR Gate, and their stick diagram. Lol...
- I was asked to draw the layout of the inverter. How will the layout of the Inverter look like? Basically, he wanted to know how much you are familiar with the layout. *Remember the VLSI Layout demo class on Mentor graphics.*

Comment:

The interviewer, in the 2nd round, was from Taiwan. He kept some words already written on the paper and whenever he felt the need to pronounce them he used to show us those words. It was due to his accent. Many of us were unable to understand his accent.

5. **Max Linear**

This company came for multiple profiles, and one could write the test at max for two profiles. I opted for ASIC Design & PnR profiles. Question papers were exactly the same as it was in the previous year, and luckily we had it. I got selected for both profiles. I was last to be interviewed, hence had relatively less chance to get selected because they were in a hurry. Now, questions…

- Explain setup and hold time and don't tell me definitions? *You can get a feeling of setup and hold time from here (http://www.vlsi-expert.com/2011/04/static-timing-analysis-sta-basic-part3a.html).*
- How will you minimize the delay in the NAND and NOR gate? *Qureshi Sir must have taught you.*
- Some questions were based on the "switching activities" topic.
- He drew some RC circuit and asked to find the delay at some node. *Elmore delay formula.*
- What is EM and how will you minimize the effect?
- Buffer depth calculation based question - One of the favorites questions of the interviewer, understand it properly.
- Write Verilog code for the given circuit. I am unable to remember what circuit was.
- I had been given a problem statement and was asked to write the C++ code. *It was an easy one.*
- He asked about my BTech. college and how did you get admission into that? *I have done my graduation from a tier-3 college. He wasn't happy with that.*

Comment:

Don't need to worry if you have done your BTech from such a college because this is the only interview where I was asked about my BTech. College.

6. **Analog Devices Incorporation**

   In the written test, the paper was exactly the same as the previous one. I will later attach the paper in the same doc. There were two sections in the paper that comprise VLSI and Digital signal processing sections.

   There were two rounds of technical interviews, and questions were highly conceptual. Basically, they wanted to check our problem-solving skills, not the final answer. They were helpful in nature.

   - How will you design the Fingerprint sensors and its algorithm to identify the registered finger? We *had an hour of* ***discussion*** *on design. This was the only question we had been asked in the 1st round of interviews.*
   - There is a memory matrix, each element of the matrix represents 1 bit, which has 8 columns(= 1Byte data) and 7 rows. Memory is loaded with four different 1Byte data in which three 1 Byte data are loaded two times and fourth 1Byte data has no pair that is loaded one time. How will you identify which data was loaded single times? *Let me know if you don't know the answer.*
   - Draw the state diagram for the mod 7 division.
   - *There were two more questions, will add as soon as I recall them.*

Comment:

In the written test, I had solved questions only from the VLSI section. I thought that they would qualify the students on the basis of the total marks obtained. But I failed here, during the interview they told me that you didn't solve any question from the digital signal processing section. So try to solve at least one or two questions from each section.

7. **Redpine Signals**

   Profiles: Digital ASIC design, RF & Microwaves, Analog IC design

   We, somehow, arranged the written test paper of the previous year and luckily it went exactly the same in our time also.

   During the interview, they asked easy questions that one could easily answer.

   - Draw the AC equivalent circuit of the NMOS and PMOS?
   - Find input resistance and voltage gain of the Common source NMOS amplifier? *Analog VLSI Book*

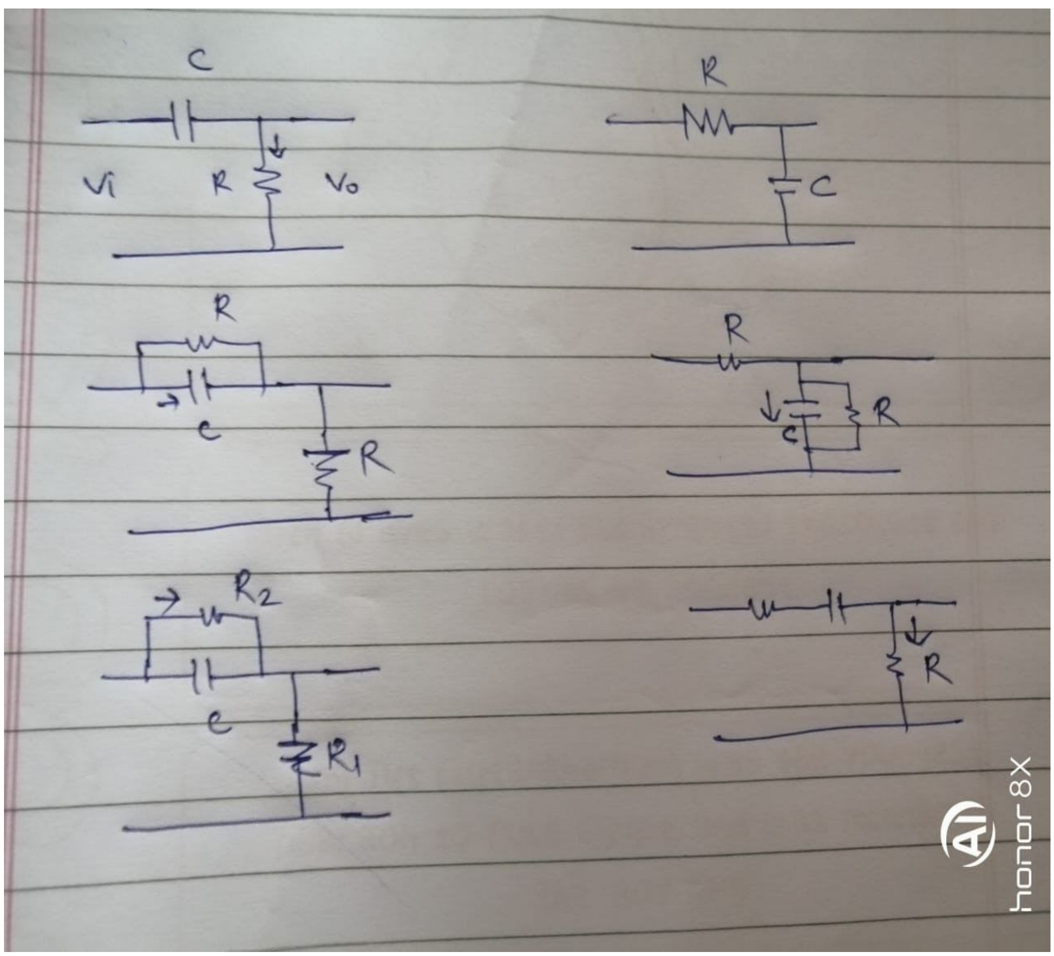
- Draw the PTAT circuit, and why and where do we use it? *Analog VLSI Book*
- The transconductance of MOSFET, Explain? *Analog VLSI Book*
- *I was given the MOS circuit and was asked to find the gain?*

8. **SignalChip**

This was my **most favorite** interview. It lasted around 2+1 hours. Interviewers were former **Texas Instruments(TI)** employees. There were two rounds.

During the first Round:

- Draw the approximate waveform of the current/voltage through/across the R and C in the given circuit. I was asked to draw waveforms **without using any formulae,** not even charging or discharging formula of the capacitor. What I could use was my **common sense**. That was the only thing I was allowed. He drew all the possible combinations of the RC circuits, with little modifications, one after another. Below are some of them.



- Find out the equivalent resistance across the diagonal of the cube, both the end of an edge of the cube and face diagonal of the cube. *He didn’t want direct answers.*
- Op-amp? Solve Practical op-amp problems.